9	**(a)**	e.g. no time delay between illumination and emission max. (kinetic) energy of electron dependent on frequency max. (kinetic) energy of electron independent of intensity rate of emission of electrons dependent on/proportional to intensity (*any three separate statements, one mark each, maximum 3*)		B3	[3]
	(b)	**(i)**	(photon) interaction with electron may be below surface	B1	
			energy required to bring electron to surface	B1	[2]

(ii) **1.** threshold frequency = 5.8×10^{14} Hz A1 [1]

2. $\Phi = hf_0$ C1

$= 6.63 \times 10^{-34} \times 5.8 \times 10^{14}$

$= 3.84 \times 10^{-19}$ (J) C1

$= (3.84 \times 10^{-19})/(1.6 \times 10^{-19})$

$= 2.4$ eV A1 [3]

or

$hf = \Phi + E_{MAX}$ (C1)

chooses point on line and substitutes values E_{MAX}, f and h into equation with the units of the hf term converted from J to eV (C1)

$\Phi = 2.4$ eV (A1)